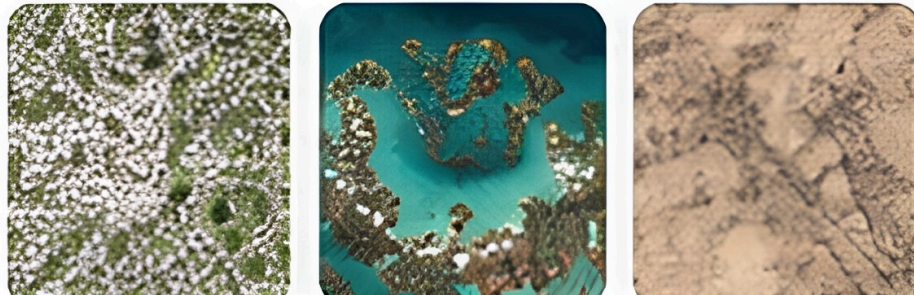


1

Georeferenced RS Samples

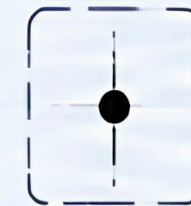
m-EuroSAT, m-BigEarthNet, m-SA-Crop-Type, MMEarth20K, ...



 Lat/Lon; Longitude (Georeferenced)

2

Biome Labeling Pipeline

 Patch Centroid

Spatial Join with Terrestrial Ecoregion Map (v6.5)

 16-Sec Ecoregional Biome

3

Two-Level Grouping

16 Fine Terrestrial Biomes

1

2

3

5

6

7

8

9

9

10

11

12

16

16

 1. Associational High-Biomes

 2. High-Amplitude Macrobiomes

 3. Transitional Interbiomes and Seams

 4. Cryospheric & Desert-Centric

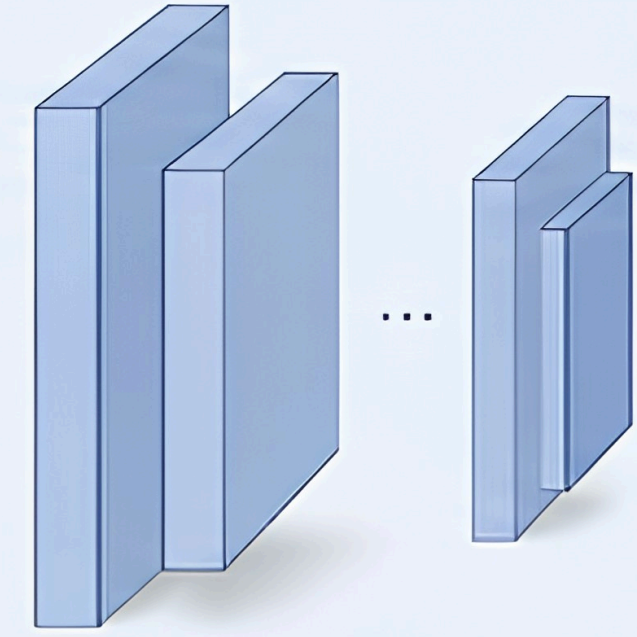
 5. Arid and Warm-adapted

 6. Hydrospheric/ Wet-dominated

4

Frozen RSFM Encoder (Prithvi-EO-2.0)

 frozen



No backpropagation when probing or debiasing

5

Linear Probing + Group-wise Evaluation

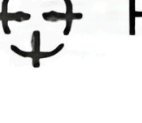
 Frozen Features (x)

 Classification Head

 Supervision Head

 Worst-Group per accuracy

 Mean-Group

 HPM

 EPM

 BPM

6

Biome-Aware Debiasing

 DBR (Dynamic)

 **BOLP (Ours)**
Representation-level (closed-form)